

Dependency Failure Handling Verification

Purpose

This test was performed to verify how the TEAVS–ADCRS integration endpoint behaves when its RabbitMQ dependency is unavailable.

The endpoint under test was:

POST /api/ingestion/core

RabbitMQ is required for forwarding accepted core integration requests to the downstream data-ingestion and model-processing workflow.

Test Method

A valid ingestion_service JWT and valid TEAVS–ADCRS payload were prepared.

RabbitMQ was then deliberately stopped in the local Docker environment before submitting the request.

Result

With RabbitMQ unavailable, the request returned:

HTTP 500 Internal Server Error

with the response:

Error integrating core model

Assessment

PASS

The endpoint did not return HTTP 202 when the integration queue was unavailable.

This confirms that the API Gateway detects the RabbitMQ failure and reports an error instead of incorrectly indicating that the integration request was accepted.

RabbitMQ was restarted immediately after the test and the API Gateway was restarted to restore the normal integration environment.

Evidence

EVID-FAIL-01 – RabbitMQ dependency failure

```
● (base) abdulrehman@Abduls-Air backend % export INGESTION_TOKEN=$(curl -s -X POST http://localhost:3001/api/users/auth/login \
-H "Content-Type: application/json" \
Show Command Actions

Command executed 1 min ago and took 1.924s
t((json.load(sys.stdin)["access_token"]))'
● (base) abdulrehman@Abduls-Air backend % docker compose stop rabbitmq
[+] Stopping 1/1
  ✓ Container phoenix-rabbitmq Stopped
● (base) abdulrehman@Abduls-Air backend % curl -i -X POST http://localhost:3001/api/ingestion/core \
-H "Authorization: Bearer $INGESTION_TOKEN" \
-H "Content-Type: application/json" \
-d '{
  "url": "https://example.com/failure-test",
  "text": "Failure handling test",
  "timestamp": "2026-09-27T09:15:00+10:00",
  "hazard_type": "phishing",
  "hazard_severity": 4,
  "hazard_timestamp": "2026-09-27T09:14:00+10:00",
  "hazard_location": "VIC",
  "hazard_status": "active",
  "alert_level": "critical",
  "source": "TEAVS"
}'
HTTP/1.1 500 Internal Server Error
X-Powered-By: Express
Access-Control-Allow-Origin: *
Content-Type: application/json; charset=utf-8
Content-Length: 55
ETag: W/"37-88UUPRK+9r8vDioXggpMj58m38s"
Date: Sat, 26 Sep 2026 23:21:19 GMT
Connection: keep-alive
Keep-Alive: timeout=5

{"status":500,"message":"Error integrating core model"}
● (base) abdulrehman@Abduls-Air backend % docker compose start rabbitmq
docker compose restart api-gateway
[+] Running 1/1
  ✓ Container phoenix-rabbitmq Started
[+] Restarting 1/1
  ✓ Container api-gateway Started
○ (base) abdulrehman@Abduls-Air backend %
```

Controlled RabbitMQ outage test showing HTTP 500 returned by the TEAVS–ADCRS integration endpoint when the messaging dependency was unavailable.